

Name: ANSWERSTime: 30 mins
/ 35 marks1 x A4 page of hand written notes (one sided)
Scientific Calculator

Part A – Multiple Choice – Circle only 1 answer

6 marks

Question 1

If a fair 6-sided die is rolled, the chance of getting a 5 is?

☒ A. $\frac{1}{6}$

B. $\frac{5}{6}$

C. 1 : 6

D. $\frac{6}{5}$

Question 2

If a fair 6-sided die is rolled, what is the chance of getting less than 4?

A. $\frac{1}{6}$

☒ B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. 4 : 6

Question 3

A bowl contains 30 identical balls numbered 1 to 30. A ball is drawn at random from the bowl. The probability that the number on the ball is a multiple of 4 is

A. $\frac{1}{30}$

B. $\frac{2}{15}$

☒ C. $\frac{7}{30}$

D. $\frac{1}{4}$

Question 4

A coin is tossed three times. What is the possibility of tossing two tails in a row?

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{1}{8}$

D. $\frac{1}{2}$

Question 5

54 students were asked how they got to school

18 walked

7 rode a scooter

15 came by car

14 used public transport

What is the relative frequency of students that used public transport?

A. 0.23

B. 0.24

C. 0.25

D. 0.26

Question 6

The colour of 20 cars passing a school were recorded over a short period of time and summarised in the frequency table below

Colour	White	Black	Blue	Red
Frequency	4	9	5	2

If the colour of 100 cars were recorded what would be the expected number of blue cars?

A. 5

B. 15

C. 25

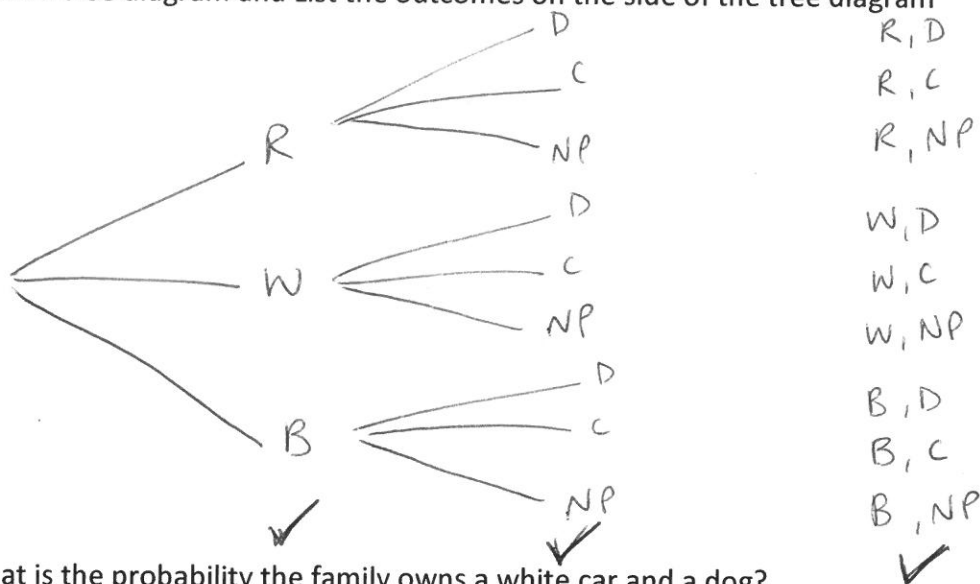
D. 50

Question 7

[6 marks]

A family has an equal chance of owning a red car, a white car or a black car. At the same time each family has an equal chance of owning a dog, a cat or no pets.

- a) Draw a tree diagram and List the outcomes on the side of the tree diagram



- b) What is the probability the family owns a white car and a dog?

$$\frac{1}{9} \quad \checkmark$$

- c) What is the probability the family has a pet and a black car?

$$\frac{2}{9} \quad \checkmark$$

- d) What is the probability the family owns a red car?

$$\frac{3}{9} \quad \checkmark$$

Question 8

[3 marks]

A die was rolled 30 times. The table below shows the frequency of each number rolled.

Die face	1	2	3	4	5	6
Frequency	7	3	6	5	6	3

- a) What is the experimental probability of rolling a 5?

$$\frac{6}{30} \quad \checkmark$$

- b) If this experiment was repeated, and the dice was rolled 100 times, what would you predict the probability would be? Give a reason

experimental probability becomes closer
to theoretical probability

Question 9

[4 marks]

Determine the following theoretical probabilities when rolling a six-sided die and tossing a coin.

(Hint: choose an appropriate sample space)

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

- a) P (a 5 and a H) $\frac{1}{12}$ ✓
- b) P (a 5 or a H) $\frac{7}{12}$ ✓
- c) P (even number and a H) $\frac{3}{12}$ ✓
- d) P (T) $\frac{6}{12}$ ✓

Question 10

[8 marks]

Three marbles (red, blue and yellow) were placed in a box. 2 marbles are removed one after the other. The colour of the first marble is noted then replaced.

- a) Complete the following 2-way table listing all the possible outcomes

		2 nd marble		
		R	B	Y
1 st marble	R	RR	RB	RY
	B	BR	BB	BY
	Y	YR	YB	YY

→ be mindful of order

- b) What is the total number of possible outcomes? 9 ✓
- c) P(both same colour) = $\frac{3}{9}$ ✓
- d) P(green) 0 ✓
- e) P(Red or blue) = $\frac{8}{9}$ ✓
- f) P(Red and blue) $\frac{2}{9}$ ✓

- g) If the first marble was NOT replaced, what would the P(Red and Blue) be? Justify your answer

$$\frac{2}{6} \checkmark$$

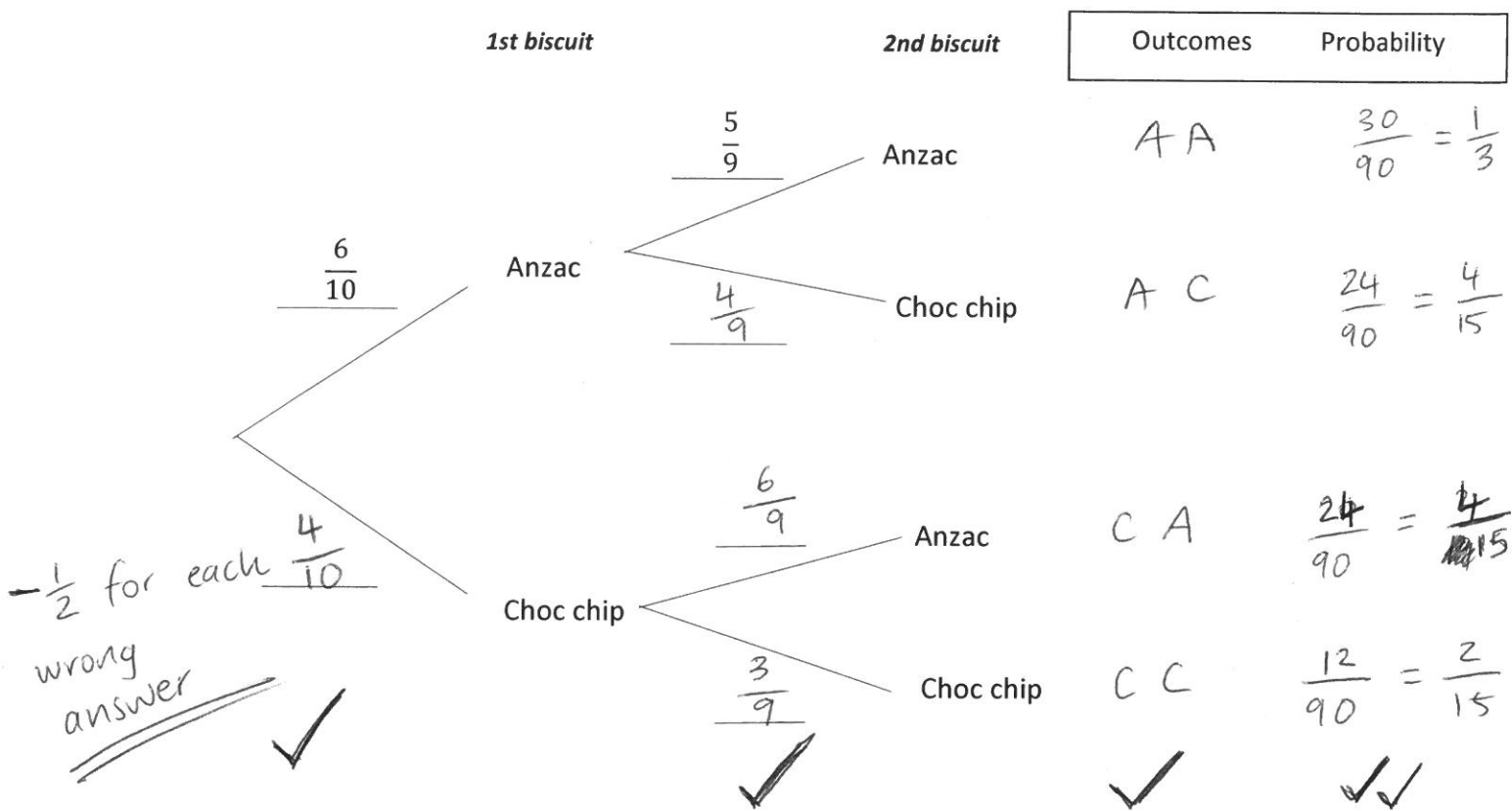
explained ✓

Question 11

[8 marks]

Jasmine has 10 biscuits in a tin (6 anzac and 4 choc chip). She chooses 2 biscuits at random to eat with her cup of tea before bed.

Complete the following tree diagram, list the outcomes and the probabilities



- a) What is the probability that Jasmine eats 2 choc chip biscuits?

$$\frac{12}{90} = \frac{2}{15} \quad \checkmark$$

- b) What is the probability that Jasmine has an Anzac biscuit first?

$$\frac{6}{10} \quad \checkmark$$

- c) What is the probability that Jasmine eats 2 of the same type of biscuit?

$$\frac{30}{90} + \frac{12}{90} = \frac{42}{90} \text{ or } \frac{7}{15} \quad \checkmark$$

